

Assignment - 5

Implement TCP communication over IEEE 802.11n wireless LAN with MPDU aggregation enabled and analyze the performance of different TCP variants under varying simulation parameters.

- Simulation Setup:
 - o Network Topology
 - One HT Station (source) transmitting TCP packets to an Access Point (sink).
 - o Link Parameters
 - Application Data Rate: 100 Mbps.
 - Propagation Delay Model: ConstantSpeedPropagationDelayModel.
 - Propagation Loss Model: FriisPropagationLossModel.
 - o Variable Parameters

- Performance Metric:
Average Throughput = (Total Received packets * 8)/ (1e6 * simulation time).

- Tasks:
 - o Simulate for multiple TCP variants (Tahoe, Reno, NewReno, Cubic, etc.).
 - o Vary simulation time and payload size and record throughput results.
 - o Analyze the results by comparing the throughput of different TCP variants.
 - o Plot graphs (using Gnuplot):
 - Throughput vs. Payload Size.
 - Throughput vs. Simulation Time.